

$$xk + y = 1$$

$$kx + y = -1$$

$$\Rightarrow \begin{bmatrix} k & 1 \\ 1 & k \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow R_2 \rightarrow kR_2 - R_1$$

$$\Rightarrow \begin{bmatrix} k & 1 \\ 0 & k^2 - 1 \end{bmatrix} = \begin{bmatrix} 1 \\ -(k+1) \end{bmatrix}$$

(Rank = 2)

unique solution $\Rightarrow k^2 - 1 \neq 0$

$$\Rightarrow k \neq \pm 1$$

$$\rightarrow \text{Infinite } k^2 = \pm 1 \quad \&\& \quad -(k+1) = 0$$

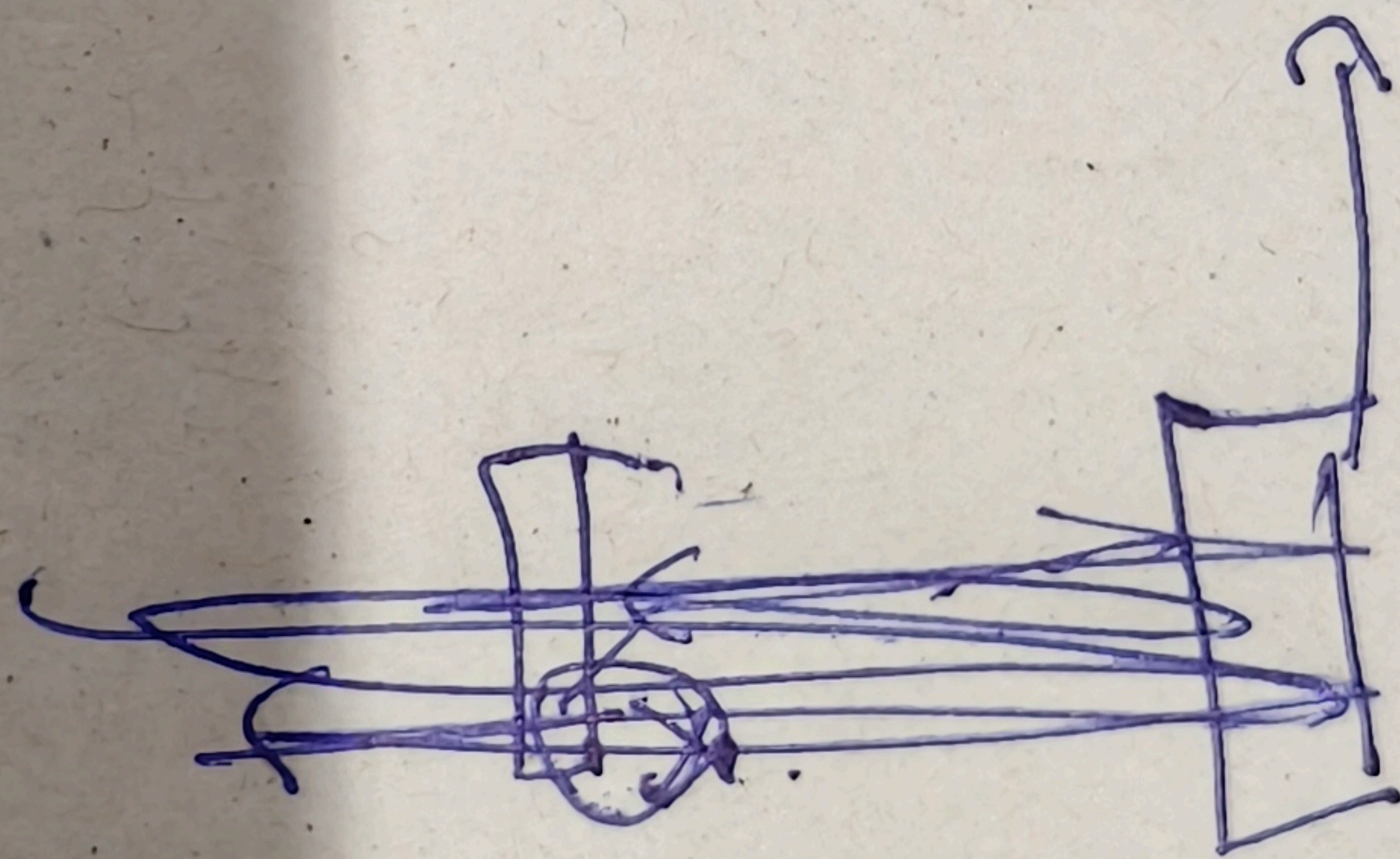
$$\Rightarrow k = -1,$$

$$\Rightarrow k = +1,$$

$\Rightarrow$ No soln,

$$\text{as } 0 = \cancel{(k+1)}$$

$$x = -2$$



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